



Motivation

- Predicting cognitive states such as mind wandering during learning is important for adaptive educational technologies
- Video-based mind-wandering detection is promising → continuous and non-intrusive, but progress is limited by data scarcity
- Prior work leveraged pretrained emotion recognition model – transfer learning to MW pred → ER models may provide useful visual representations for MW detection
- Do more recent and stronger emotion-recognition foundation models improve frozen-transfer performance for video-based mind-wandering detection?

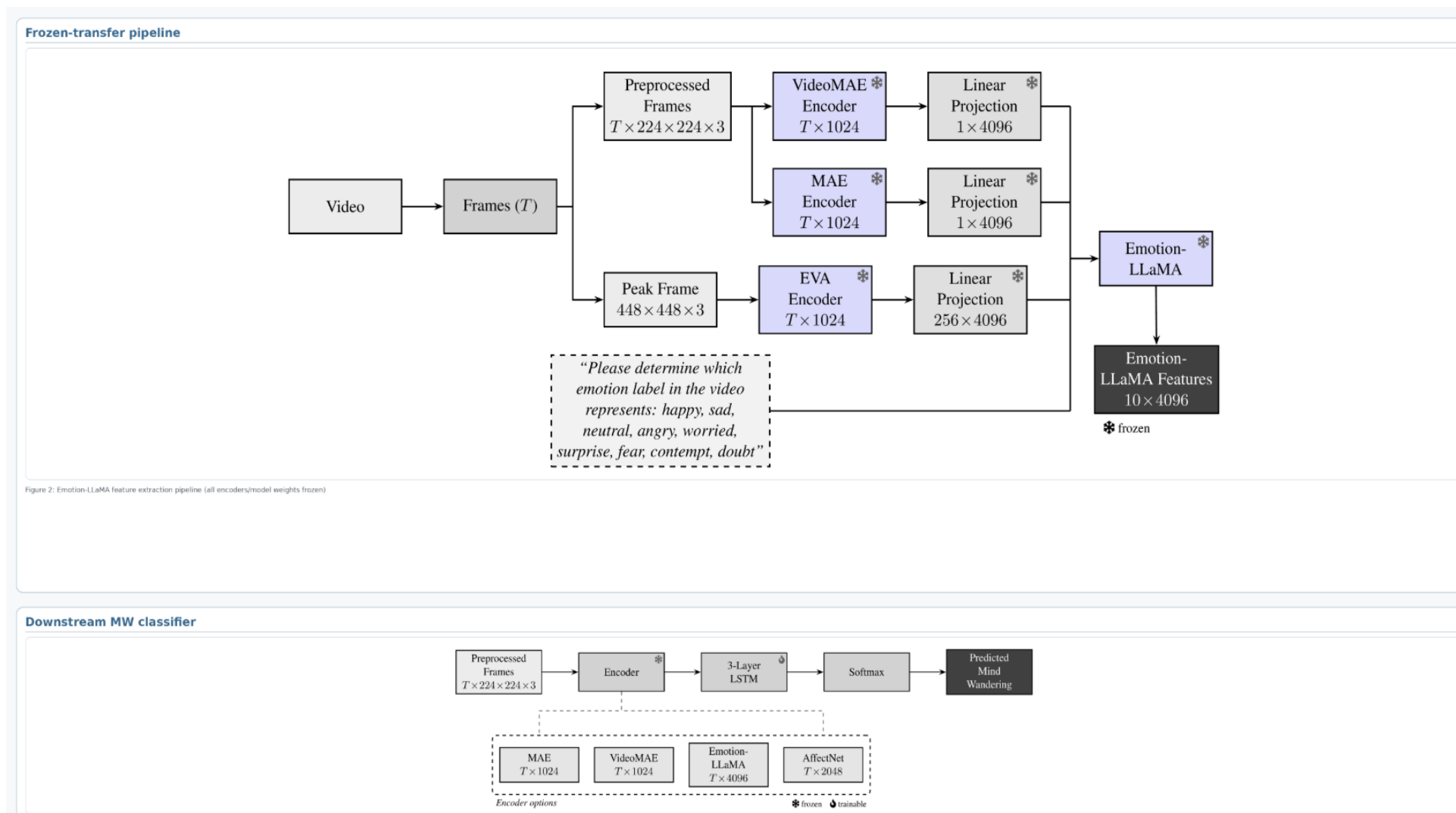
Experiments

- Building on SOTA transfer-learning pipeline, keep downstream MW classifier fixed
 - swap encoder w. newer pretrained ER
- Baseline: AffectNet-pretrained ResNet50
- Newer encoders: MAE, VideoMAE, Emotion-LLaMA

Evaluation

- Metrics: F1, Precision, Recall, and ROC-AUC
- Analysis: How each encoder fails
 - prediction confidence and score separability
 - FP vs. FN error profiles
 - shared vs. encoder-specific failures
 - predicted emotion labels vs. MW errors

Methodology



Results

Main result: stronger ER models do not win

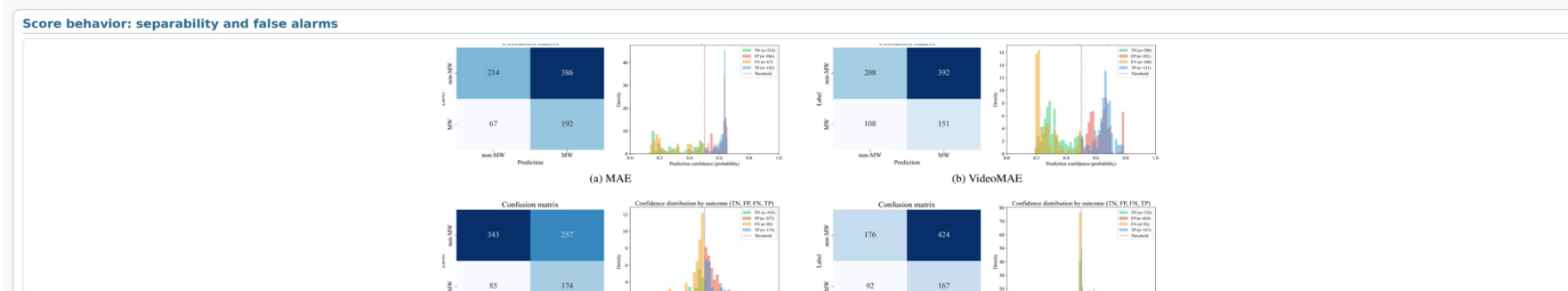
Encoder	F1	Precision	Recall	ROC-AUC
Affectnet	0.4622	0.3657	0.6834	0.6249
MAE	0.4283	0.3371	0.6492	0.5319
VideoMAE	0.4096	0.3128	0.6354	0.5254
Emotion-LLaMA	0.4232	0.2977	0.7425	0.5002

AffectNet
Best avg. F1 + ROC-AUC
F1 0.462 | AUC 0.625

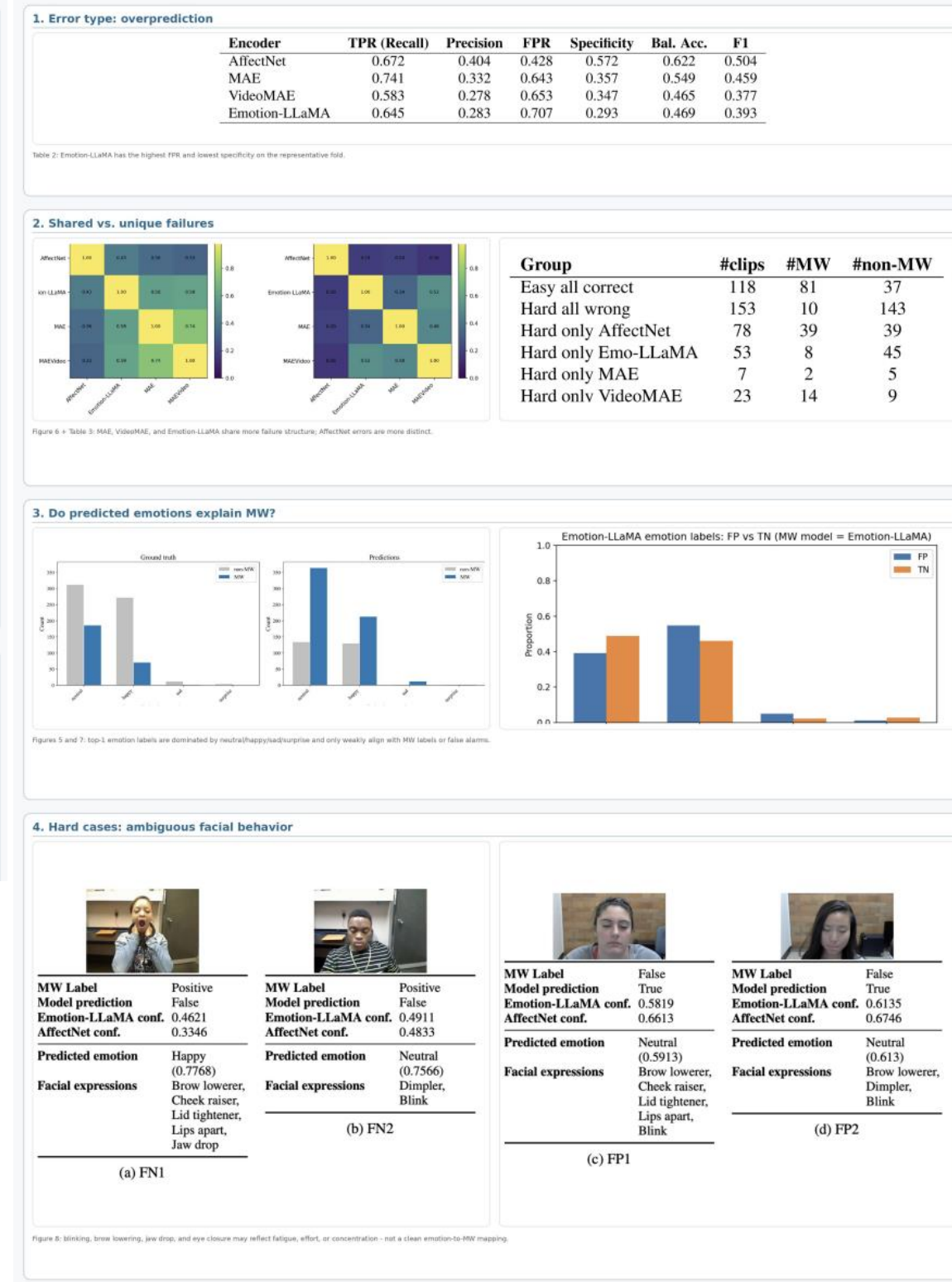
Emotion-LLaMA
Highest Recall, low Precision
Recall 0.743 | Prec. 0.298

Interpretation
Learning from MW clips is difficult, many false alarms
MW overprediction

Takeaway The newer Emotion-LLaMA-based representations do not surpass the AffectNet baseline, despite greater architectural sophistication and stronger ER benchmark performance.



Analysis



Outlook

SOTA ER performance → poorer MW prediction

- emotion descriptors align weakly with MW-relevant behavior, weaker class separability, overpredict.
- Move beyond emotion-centric transfer → task-